



BlueField Operation

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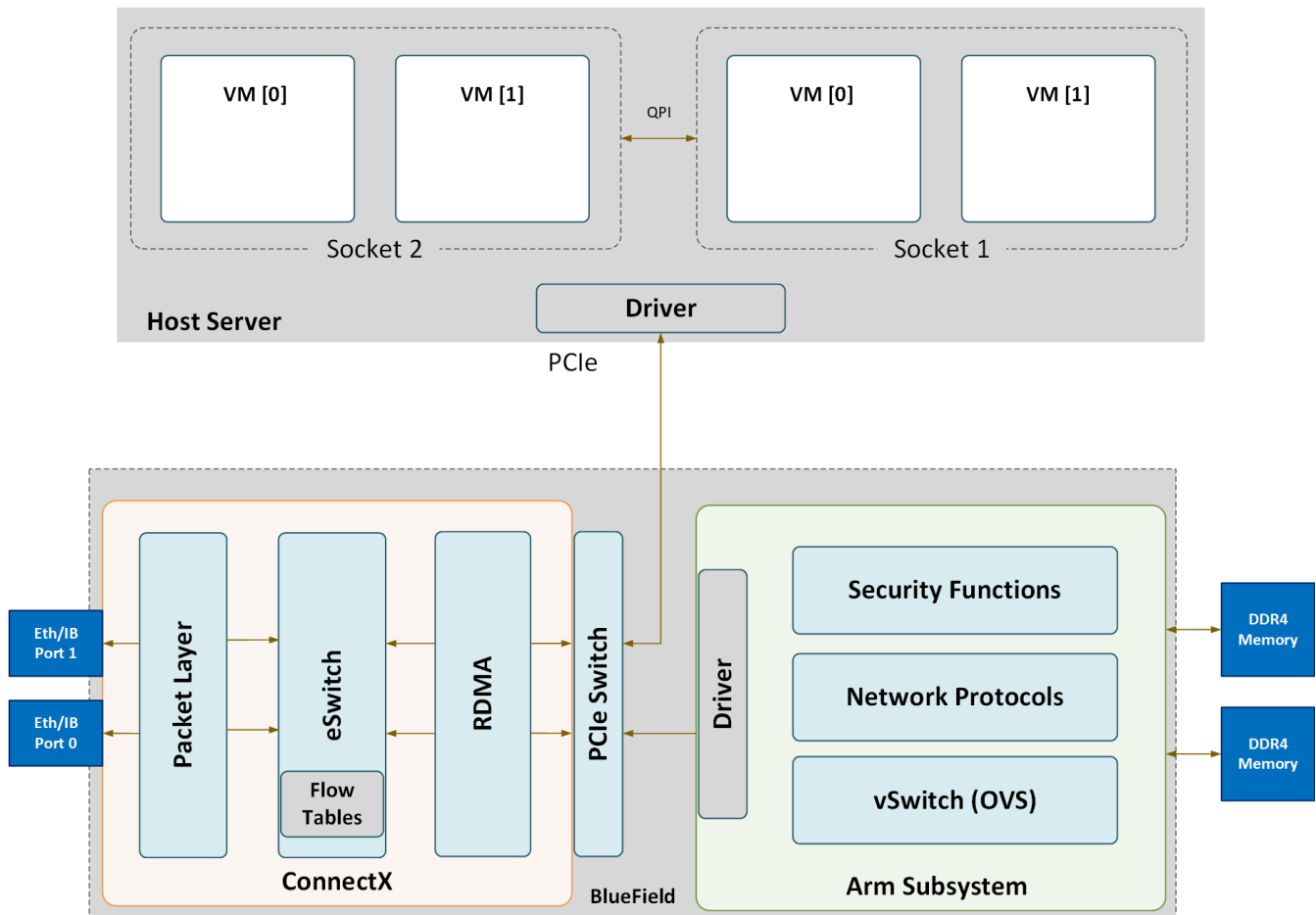
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The NVIDIA® BlueField® networking platform family delivers the flexibility to accelerate a range of applications while leveraging ConnectX-based network controllers hardware-based offloads with unmatched scalability, performance, and efficiency.

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Functional Diagram

The following is a functional diagram of the NVIDIA® BlueField®-2 DPU.



For each network port of the BlueField networking platform (DPU or SuperNIC), there are 2 physical PCIe networking functions exposed:

- To the embedded Arm subsystem
- To the host over PCIe

Note

Different functions have different default grace period values during which functions can recover from/handle a single fatal error:

- ECPFs have a graceful period of 3 minutes
- PFs have a graceful period of 1 minute
- VFs/SFs have a graceful period of 30 seconds

The mlx5 drivers and their corresponding software stacks must be loaded on both hosts (Arm and the host server). The OS running on each one of the hosts would probe the drivers. BlueField-2 network interfaces are compatible with NVIDIA® ConnectX®-6 and higher. BlueField-3 network interfaces are compatible with ConnectX-7 and higher.

The same network drivers are used both for BlueField and the ConnectX NIC family.

BlueField SR-IOV

The NVIDIA® BlueField® SR-IOV solution is based on asymmetric VF and enables per-ECPF and per PF control over number of VF allocation .

ECPF VFs are intended to be used in switchdev mode. Like SFs and host VFs, ECPF VFs have a representor. Representor naming for ECPF VFs start after the host VFs. For example, if the host has 32 VFs enabled, then the host VF representors are named `pf0vf0` - `pf0vf31`, and the Arm representors continue at `pf0vf32` onward.

To enable BlueField SR-IOV, apply the following configuration in the BlueField OS:

```
mlxconfig -d 03:00.0 -y s PF_NUM_OF_VF_VALID=1
```

Note

Once `PF_NUM_OF_VF_VALID` is set, the `NUM_OF_VFS` `mlxconfig` option is not relevant and the user must set `PF_NUM_OF_VF` for each host and EC function. It is recommended for the number of VFs for each ECPF and each host PF be the same.

The BlueField should now support setting asymmetric VF configuration per port.

The following are examples for configuring the number of VFs per port:

1. In the BlueField, issue the following commands to configure 32 VFs per port:

```
bf> mlxconfig -d 03:00.0 -y s PF_NUM_OF_VF=32
bf> mlxconfig -d 03:00.1 -y s PF_NUM_OF_VF=32
```

Note

The BlueField ECPF driver in the BlueField's Arm OS limits the number of VFs it supports to 32 per port.

2. In the host OS, issue the following commands to configure up to 126 VFs per port:

```
host> mlxconfig -d 03:00.0 -y s PF_NUM_OF_VF=126  
host> mlxconfig -d 03:00.1 -y s PF_NUM_OF_VF=126
```

3. Perform a [BlueField system reboot](#) for the `mlxconfig` settings to take effect.

4. Create ECPF VFs:

```
echo 1 > /sys/class/net/p0/device/sriov_numvfs
```

Note

BlueField SR-IOV VFs do not support the following legacy SR-IOV functionalities:

- Virtual switch tagging (VF VLAN)
- Spoof check
- VF trust
- VF rate

fTPM over OP-TEE

Note

fTPM over OP-TEE is supported on NVIDIA® BlueField®-3 DPUs and higher only on host OS Ubuntu 22.04 or Oracle Linux.

The Trusted Computing Group (TCG) is responsible for the specifications governing the trusted platform module (TPM). In many systems, the TPM provides integrity measurements, health checks and authentication services.

Attributes of a TPM:

- Support for bulk (symmetric) encryption in the platform
- High quality random numbers
- Cryptographic services
- Protected persistent store for small amounts of data, sticky bits, monotonic counters, and extendible registers
- Protected pseudo-persistent store for unlimited amounts of keys and data
- Extensive choice of authorization methods to access protected keys and data
- Platform identities
- Support for platform privacy
- Signing and verifying digital signatures
- Certifying the properties of keys and data
- Auditing the usage of keys and data

With TPM 2.0., the TCG creates a library specification describing all the commands or features that could be implemented and may be necessary in servers, laptops, or embedded systems. Each platform can select the features needed and the level of security or assurance required. This flexibility allows the newest TPMs to be applied to many embedded applications.

Firmware TPM (fTPM) is implemented in protected software. The code runs on the main CPU so that a separate chip is not required. While running like any other program, the code is in a protected execution environment called a trusted execution environment (TEE) which is separate from the rest of the programs running on the CPU. By doing this, secrets (e.g., private keys perhaps needed by the TPM but should not be accessed by others) can be kept in the TEE creating a more secure environment.

Info

fTPM provides similar functionality to a chip-based TPM, but does not require extra hardware. It complies with the official TCG reference implementation of the [TPM 2.0 specification](#). The source code of this implementation is located [here](#).

Info

fTPM fully supports [TPM2 Tools](#) and the TCG TPM2 Software Stack ([TSS](#)).

Characteristics of an fTPM:

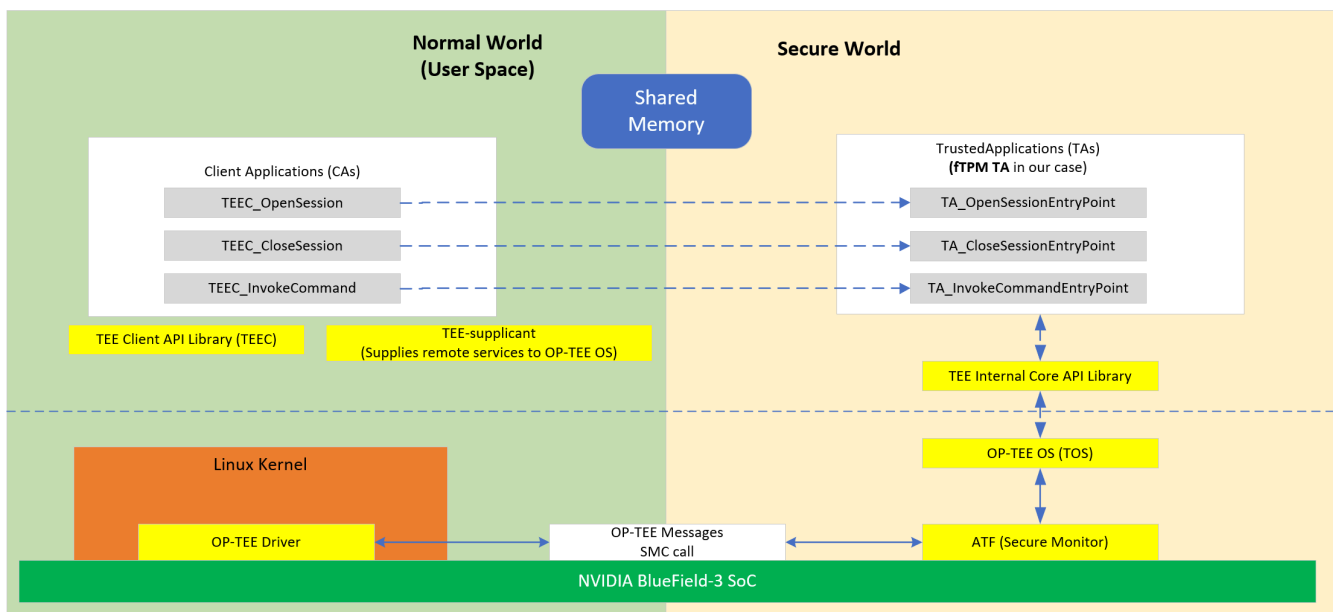
- Emulated TPM using an isolated hardware environment
- Executes in an open-source trusted execution environment (OP-TEE)
- fTPM trusted application (TA) is part of the OP-TEE binary. This allows early access on bootup, runs only in secure DRAM.

Info

Currently, the only TA supported is fTPM.

- fTPM is not a task waiting to be woken up. It only executes when TPM primitives are forwarded to it from the user space. It is guaranteed shielded execution via the TEE OS and, when invoked via the TEE Dispatcher, runs to completion.

The fTPM TA is the only TA BlueField-3 currently supports. Any TA loaded by OP-TEE must be signed (signing done externally) and then authenticated by OP-TEE before being allowed to load and execute.

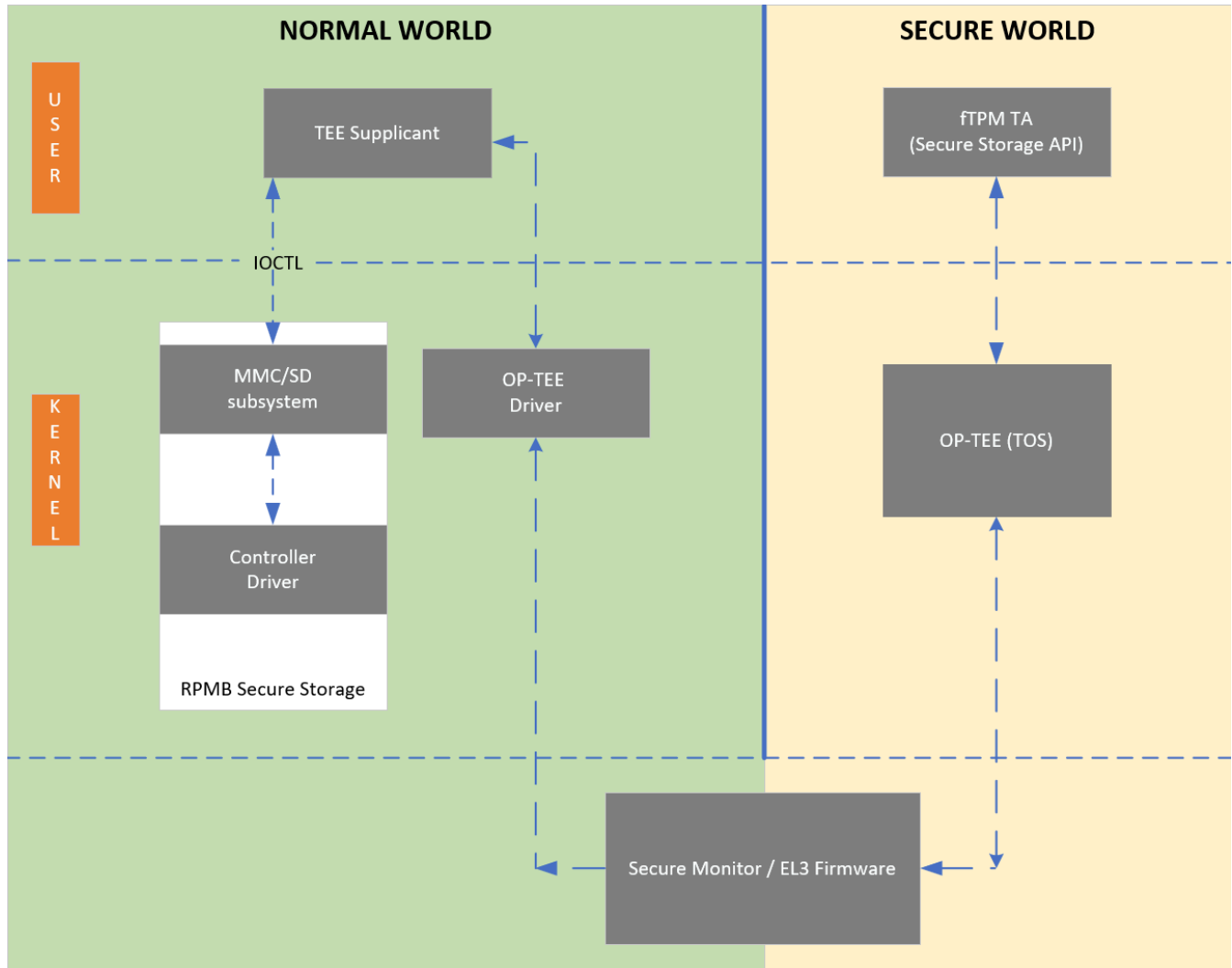


A replay-protected memory block (RPMB) is provided as a means for a system to store data to the specific memory area in an authenticated and replay-protected manner, making it readable and writable only after a successful authentication read/write accesses. The RPMB is a dedicated partition available on the eMMC, which makes it possible to store and retrieve data with integrity and authenticity support. A signed access to an RPMB is supported by first programming authentication key information to the eMMC memory (shared secret). The RPMB authentication key is programmed into BlueField at manufacturing time.

Info

RPMB features a 4MB partition secure storage for BlueField-3.

There is no eMMC controller driver in OP-TEE. All device operations have to go through the normal world via the TEE-suppliant daemon, which relies on the Linux kernel's ioctl interface to access the device. All writes to the RPMB are atomic, authenticated, and encrypted. The RPMB partition stores data in an authenticated, replay-protected manner, making it a perfect complement to fTPM for storing and protecting data.



Enabling OP-TEE on BlueField-3

Enable OP-TEE in the UEFI menu:

1. ESC into the UEFI on BlueField boot.
2. Navigate to Device Manager > System Configuration.
3. Check "Enable OP-TEE".
4. Save the change and reset/reboot.

5. Upon reboot OP-TEE is enabled.

Note

OP-TEE is essentially dormant (does not have an OS scheduler) and reacts to external inputs.

Verifying BlueField-3 is Running OP-TEE

Users can see the OP-TEE version during BlueField-3 boot:

```
Nvidia BlueField-3 rev1 BL1 V1.0
INFO: psc_supervisor_init.
INFO: psc_irq_init...
INFO: force_crs_enable=0 pcr.lock0 = 0, time = 111291
INFO: enter_idle task.
NOTICE: Running as 9009D3B400ENEA system
NOTICE: BL2: v2.2(release):4.5.0-16-g2bd9b06e2-dirty
NOTICE: BL2: Built : 15:43:42, Sep 7 2023
NOTICE: BL2 built for hw (ver 2)
NOTICE: # Finished initializing DDR MSS1
NOTICE: DDR POST passed.
INFO: mailbox rx: channel = 2, code = 0x43544c44
NOTICE: BL31: v2.2(release):4.5.0-16-g2bd9b06e2-dirty
NOTICE: BL31: Built : 15:43:44, Sep 7 2023
NOTICE: BL31 built for hw (ver 2), lifecycle Production

PTM:171288:2:0:6~
I/TC:
I/TC: OP-TEE version: 3.10.0-21-g450b24a (gcc version 8.3.0 (GCC)) #1 Sat Aug 26 11:54:32 UTC 2023 aarch64
I/TC: Primary CPU initializing
I/TC: Primary CPU switching to normal world boot
UEFI firmware (version BlueField:4.5.0-16-g0e7fa9c192-BId0 built at 20:53:10 on Sep 6 2023)
```

The following indicators should all be present if fTPM over OP-TEE is enabled:

- Check "dmesg" for the OP-TEE driver initializing

```
root@localhost ~]# dmesg | grep tee
[ 5.646578] optee: probing for conduit method.
[ 5.653282] optee: revision 3.10 (450b24ac)
[ 5.653991] optee: initialized driver
```

- Verify that the following kernel modules are loaded (running):

```
[root@localhost ~]# lsmod | grep tee
tpm_ftpm_tee          16384  0
optee                 49152  1
tee                   49152  3 optee,tpm_ftpm_tee
```

- Verify that the proper devices are created/available (4 in total):

```
[root@localhost ~]# ls -l /dev/tee*
crw----- 1 root root 234,  0 Sep  8 18:24 /dev/tee0
crw----- 1 root root 234, 16 Sep  8 18:24 /dev/teepriv0

[root@localhost ~]# ls -l /dev/tpm*
crw-rw---- 1 tss root  10,   224 Sep  8 18:24 /dev/tpm0
crw-rw---- 1 tss tss  252, 65536 Sep  8 18:24 /dev/tpmrm0
```

- Verify that the required processes are running (3 in total):

```
[root@localhost ~]# ps axu | grep tee
root          707  0.0  0.0 76208 1372 ?        Ssl  14:42
0:00 /usr/sbin/tee-supPLICant
root          715  0.0  0.0      0   0 ?        I<   14:42
0:00 [optee_bus_scan]

[root@localhost ~]# ps axu | grep tpm
root          124  0.0  0.0      0   0 ?        I<   18:24
0:00 [tpm_dev_wq]
```

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